

Analysis of Algorithm (15p)

1. what is the time complexity of the given function?

<pre>def ex1(n): count = 0 i = n while i >= 1: count += 1 i = i // 2 return count</pre>	$O(\log n)$	<pre>def ex2(n): count = 0 for i in range(n): count += ex1(n) return count</pre>	$O(n \log n)$
<pre>def ex3(n): count = 0 for i in range(n): for j in range(100): count += 1 return count</pre>	$O(n)$	<pre>def ex(n): sum = 0 for i in range(n): sum += i; for j in range(n): for k in range(n): sum += j * k</pre>	$O(n^2)$
<pre>def ex5(n): x = 0 i = n while (i > 1) : x += i % 10 i = i % 2 return x</pre>	$O(1)$		

True/ False (11p)

2. True/ False questions....

If $g(n) = 1$, the growth rate is constant and does not depend on the size of the problem	True
The address of the first node in a linked list is stored in a separate location/pointer, called head.	True
The link component of each node is a pointer.	True
A linked list is a random-access data structure such as an array.	False
We can traverse a singly linked list backward starting from the last node to the first node.	False
The elements at the bottom of the stack have been in the stack the longest.	True
Stacks can be used to print a list backward without using recursion.	True
The recursive algorithm must have three or more base cases	False
A general solution to a recursive algorithm must eventually reduce to a base case.	True
An arbitrary graph with $G(V;E)$, with $ E = V $ edges is a tree.	False. A tree has only $ V -1$ edges
Kruskal's algorithm for minimum weight spanning trees is an example of a divide-and-conquer programming algorithm.	False

Stack and Queue (4p)

3. Consider the following sequence of operations on an empty stack. `push(54); push(52); pop(); push(55); push(62); s = pop();`

Consider the following sequence of operations on an empty queue. `enqueue(21); enqueue(24); dequeue(); enqueue(28); enqueue(32); q = dequeue();`

The value of $s + q$ is _____

- A. 86
- B. 68
- C. 52
- D. 86

solution: 86

Linked list (6p, 6p)

4. The given function is going to find the sum of the values of the nodes in the singly linked list. Fill in the blank.

```
def sumOfNodes(head):  
    temp = head  
    sum = 0  
    while (-----):  
        -----  
        -----  
    return sum
```

Solution:

```
def sumOfNodes(head):  
    temp = head  
    sum = 0  
    while (temp != None):  
        sum = sum + temp.data  
        temp = temp.next  
    return sum
```

5. Consider the sumofNodes function in the previous example that you have completed. Write the function in a recursive way.

```
def sumOfNode(head):  
    if head is None:  
        return 0  
    return -----
```

solution

```
def sumOfNode(head):  
    if head is None:  
        return 0  
    return head.data + sumOfNode(head.next)
```

Recursion (4p):

6. What would be the value of combination (3,2)?

```
def combination(n,m):  
    if (m == n) or (m == 0):  
        return 1  
    else:  
        return (combination(n-1, m ) + combination(n-1, m-1))
```

C

- A. 2
- B. 6
- C. 3
- D. 4

Sorting algorithm (6p, 6p)

7. Consider the following array. [10,9,8,7,6,5,4,3,2,1]. Show how the list is sorted by the Shell Sort algorithm.

Sublist1: {10,5}, 5,9,8,7,6,10,4,3,2,1

Sublist1: {9,4}, 5,4,8,7,6,10,9,3,2,1

Sublist1: {8,3}, 5,4,3,7,6,10,9,8,2,1

Sublist1: {7,2}, 5,4,3,2,6,10,9,8,7,1

Sublist1: {6,1}, 5,4,3,2,1,10,9,8,7,6

*****5,4,3,2,1,10,9,8,7,6

Sublist1: {5,3,1,9,7},

3,4,5,2,1,10,9,8,7,6

Sublist1: {4,2,10,8,6}

*****1,2,3,4,5,6,7,8,9,10

Huffman Tree (6p)

8. A table of characters and corresponding frequencies is provided below.

Which of these is a Huffman Code for the character frequencies? Assign 0 to the left branch and 1 to the right branch.

Character	Frequency
A	45
B	13
C	12
D	16
E	9
F	5

0 101 100 111 1101 1100

- A. 'a' is 0, 'b' is 101, 'c' is 100, 'd' is 111, 'e' is 1101, 'f' is 1100
- B. 'a' is 01, 'b' is 110, 'c' is 01, 'd' is 111, 'e' is 0001, 'f' is 1101
- C. 'a' is 001, 'b' is 100, 'c' is 10, 'd' is 101, 'e' is 1100, 'f' is 1101
- D. 'a' is 0, 'b' is 101, 'c' is 100, 'd' is 01, 'e' is 0001, 'f' is 1100

A

Decode the message> FACE

Hashing (6p)

9. Use the double hash function to insert these integer keys into the hash table: 8,56,89,7,23,6

Table size is = 7

PRIME = 5

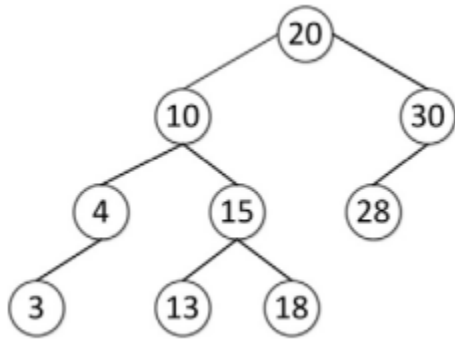
Hash1(key) = key % TABLE_SIZE

Hash2(key) = PRIME - (key % PRIME)

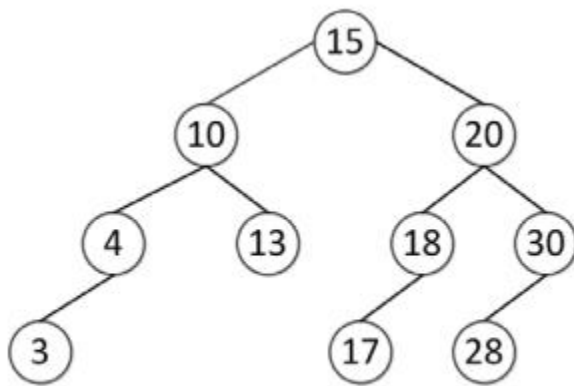
0	1	2	3	4	5	6
56	8	23	7	-	89	6

AVL Tree (6p, 6p)

10. What is result AVL tree after we insert the element 17 into the AVL tree below??



Solution:



Perform the appropriate rotation to restore the height balance of the tree. What is the level order traversal of the tree after it has been balanced?

15,10,20,4,13,18,30,3,-,-,17,-,28,-

11. Find the preorder traversal of a Binary Search Tree (BST) with the given post-order.

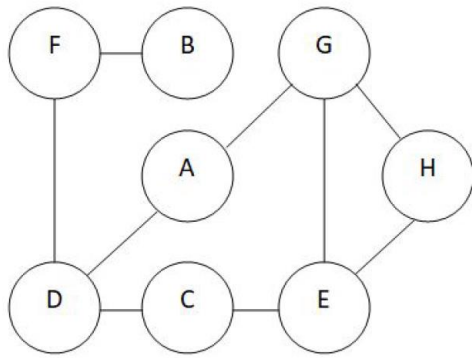
Post-order: 5,6,15,10,23,24,22,26,20

Solution:

20,10,6,5,15,26,22,24,23

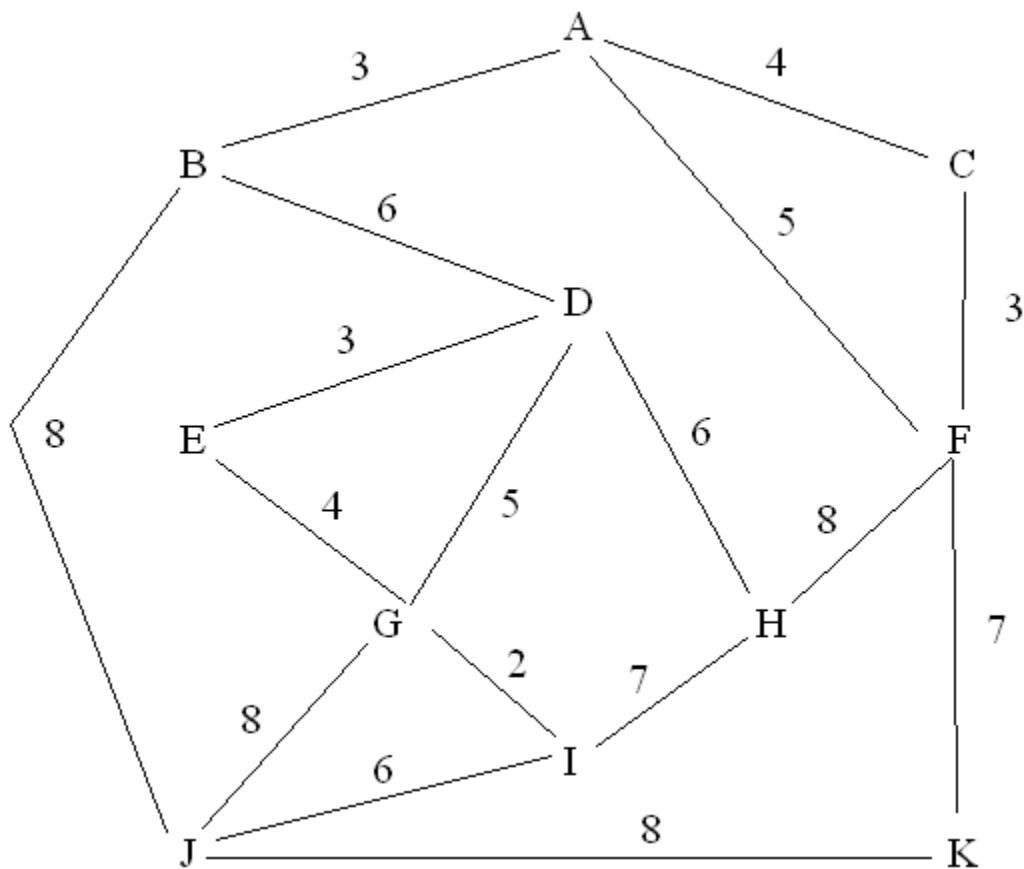
Graph (6p, 6p, 6p)

12. Run a breadth first search through the following graph starting with node A, and record the order that each node is visited. If a node has more than one neighbor, the next node to visit will be the one that comes first lexicographically.



ADGCFEHB

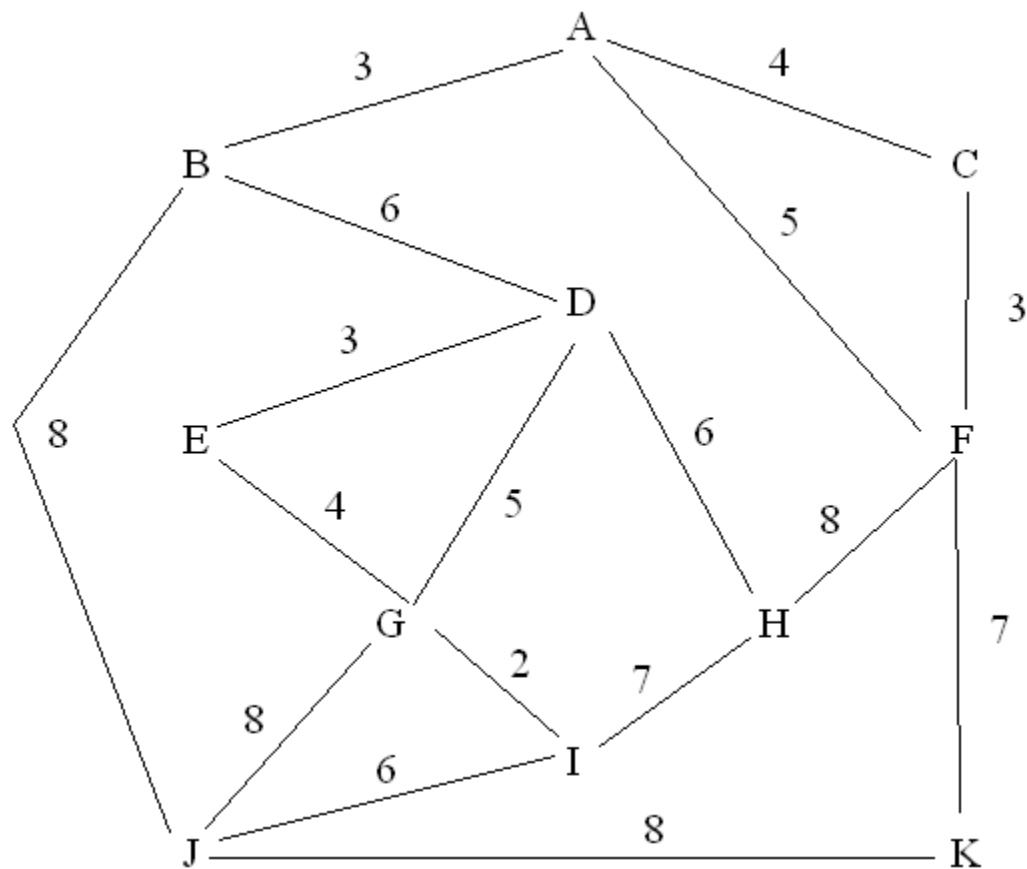
13. consider the graph in the given figure. Always visit adjacent nodes in alphabetical order. Using Dijkstra's algorithm, determine the shortest path from node A to other nodes.



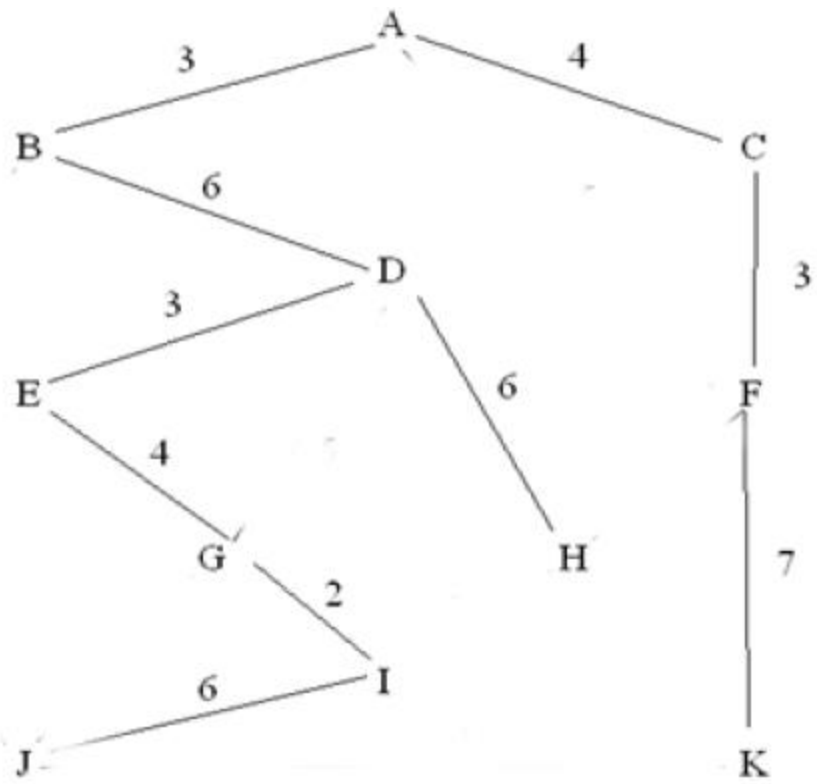
Solution:

$B > 3$
 $C > 4$
 $D > 9$
 $E > 12$
 $F > 5$
 $G > 14$
 $H > 13$
 $I > 16$
 $J > 11$
 $K > 12$

14. consider the graph in the given figure. Always visit adjacent nodes in alphabetical order. Use Prim's algorithm to derive the MST starting at node A.



Solution:



Solution: 44